UCX Cost Optimization

Projected Savings Per Month: ~$103,000* (70% reduction)

- 💰 **Delete unused Azure App Service Plan : ~ $1500**
- 💰 **Optimize Log Analytics: ~ $68,000**
- 💰 **Cosmos Db Optimization: ~$34,000**

* Based on the month of Sep, 2023 cost: $146,665.00

Priority	Recommendation	Proposed $$ savings / Months	Status	Remark
P0	Delete App plan not in use (DEV and PROD) APP Service Optimization	$1480.00	DEV and IN! plans dropped on 10/19. PROD - last one dropped on 10/31	Vanguard + Offshore
P0	**Disable profiler** App insight : ehcuspd1-ai-ucx Resource Group: pd1_rsg_cus_ucx-linux	$30,000.00	Profiler disabled on 10/18	Vanguard
P0	**Reduce RU/S from the cosmos DB** **RSG:** pd1_rsg_cus_ucx **Resource Name:** ehcuspd1cdbucxdeterminationservice	$15,000.00	DB re-sized on 10/23 .	Vanguard

			$22,800/ Month Saved	
P0	**Reduce logging levels from app code:** • 10% of your Memory was spent in LoggerExtensions.Log called from evicore.gravity.common.Service.MessageSenderService.SendAsync. We expected this value to be <2%. • 10% of your Memory was spent in SerilogLogger.Log called from evicore.gravity.common.Service.MessageSenderService.SendAsync. We expected this value to be <3%. • 7% of your Memory was spent in LoggerExtensions.Log called from evicore.gravity.common.Service.KafkaMessageConsumer.SendLagMetric. We expected this value to be <2%. • 7% of your Memory was spent in SerilogLogger.Log called from evicore.gravity.common.Service.KafkaMessageConsumer.SendLagMetric. We expected this value to be <3%. • 4% of your Memory was spent in LoggerExtensions.Log called from UCX.ACLService.Services.Observers.UcxRequestForServiceSubmittedObserver.MessageEventHandler. We expected this value to be <2%. • 4% of your Memory was spent in SerilogLogger.Log called from UCX.ACLService.Services.Observers.UcxRequestForServiceSubmittedObserver.MessageE	$30,000.00	IN Progress	Vanguard - Gravity.Common fix

	ventHandler. We expected this value to be <3%. • 6% of your Memory was spent in LoggerExtensions.Log called from evicore.gravity.common.Service.MessageConsumerService.MessageEventHandler. We expected this value to be <2%. • 6% of your Memory was spent in SerilogLogger.Log called from evicore.gravity.common.Service.MessageConsumerService.MessageEventHandler. We expected this value to be <3%.			
P1	Reduce App Service Instance count in DEV, IN and PROD [APP Service Optimization]	$23,500.00	Yet to start	probably offshore
	Approx Total Savings	**$100,000.00**		

⚠ **APP Service not in use:**

APP Name	Azure Service/Resource	RG Name	Resource Name	Action taken/ Remarks.	Saving Per Month	Remark
UCX	App Service plan	dv1_rsg_ucx-linux	eheu2dv1-asp-ucxaclmdm	Deleted on 10/19	**$202.00**	Polaris
UCX	App Service plan	dv1_rsg_ucx-linux	eheu2dv1-asp-ucxmemberhistory	Deleted on 10/19	**$577.00**	Polaris
UCX	App Service plan	dv1_rsg_ucx-linux	eheu2dv1-asp-ucxmetrics	Deleted on 10/19	**$0.00**	Polaris

UCX	App Service plan	dv1_rsg_ucx	ASP-dv1rsgucx-8d2d	Deleted on 10/19	**$115.00**	Polaris
UCX	App Service plan	dv1_rsg_ucx	ASP-dv1rsgucx-915b	Deleted on 10/19	**$189.00**	Polaris
UCX	App Service plan	in1_rsg_ucx-linux	eheu2in1-asp-ucxaclmdm	Deleted on 10/19	**$0.00**	Polaris
UCX	App Service plan	pd1_rsg_cus_ucx-linux	ehcuspd1aspucxclinicalreview	Deleted on 10/25	**$201.00**	Vanguard
UCX	App Service plan	pd1_rsg_ucx	ehcuspd1aspucx	Deleted on 10/31	**$201.00**	Vanguard

⚠ **LOG Analytics Optimization:**

	Azure Service/Resource	RG Name	Resource Name	Optimization date	Action taken/ Remarks.	Saving Per Month	Remark
1	LOG Analytics Workspace	pd1_rsg_cus_ucx-linux	ehcuspd1-law-ucx		• Changed AppTraces table storage plan from Log Analytics to Basic in DEV • Changed the retention period of data at workspace level from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV • Special handle for Application insights tables'	$60000	Wizards Deployed to Prod 11.8.

					retention period reduced from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV		
2	LOG Analytics Workspace	prod_rsg_eu2_requestrouting	eheu2prod-law-requestrouting		• Changed AppTraces table storage plan from Log Analytics to Basic in DEV • Changed the retention period of data at workspace level from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV • Special handle for Application insights tables' retention period reduced from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV	$5000	Skywalkers Deployed to Prod 11.10
3	LOG Analytics Workspace	dev_rsg_eu2_requestrouting	eheu2dev-law-requestrouting		• Changed AppTraces table storage plan from	$170	Skywalkers

					Log Analytics to Basic in DEV • Changed the retention period of data at workspace level from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV • Special handle for Application insights tables' retention period reduced from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV		Deployed to Prod 11.10
4	LOG Analytics Workspace	in1_rsg_ucx-linux	eheu2in1-law-ucx		• Changed AppTraces table storage plan from Log Analytics to Basic in DEV • Changed the retention period of data at workspace level from current 90 to 60 days in PROD & current 90 days to 30	$2000	Wizards Deployed to Prod 11.8.

					days in INTG and DEV • Special handle for Application insights tables' retention period reduced from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV		
5	LOG Analytics Workspace	dv1_rsg_ucx-linux	eheu2dv1-law-ucx		• Changed AppTraces table storage plan from Log Analytics to Basic in DEV • Changed the retention period of data at workspace level from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV • Special handle for Application insights tables' retention period reduced from current 90 to 60 days in PROD & current 90 days	$500	Wizards Deployed to Prod 11.8.

					to 30 days in INTG and DEV		
6	LOG Analytics Workspace	test_rsg_eu2_requestrouting	eheu2test-law-requestrouting		• Changed AppTraces table storage plan from Log Analytics to Basic in DEV • Changed the retention period of data at workspace level from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV • Special handle for Application insights tables' retention period reduced from current 90 to 60 days in PROD & current 90 days to 30 days in INTG and DEV	$400	Skywalkers Deployed to Prod 11.10

COSMOS DB Optimization:

1. Total RU/s Configured is very high and the normalized usage is less than 10% of allocation.
2. Reduce the RU value and programmatically increase or decrease RUs based on demand.
3. long term measure not to set no limit option.

	Azure Service/Resource	RG Name	Resource Name	Optimization date	Action taken/ Remarks.	Project cost saving per month.	Remark
1	COSMOS DB	pd1_rsg_cus_ucx	ehcuspd1cdbucxdeterminationservice		1. 3008000 RU/s Configured with less than 10% consumption.	$13,300	Vanguard
2	COSMOS DB	pd1_rsg_cus_ucx	ehcuspd1cdbucxdiagnosisservice		1. 1280000 RU/s Configured with less than 1% consumption.	$7000	Vanguard
3	COSMOS DB	pd1_rsg_cus_ucx	ehcuspd1cdbucxqualifier		1. 1280000 RU/s Configured with less than 1% consumption.	$7000	Vanguard
4	COSMOS DB	in1_rsg_eu2_ucx	eheu2in1cdbucxqualifier		1. 1168000 RU/s Configured with less than 1%	$7000	Vanguard

					consumption.		